

Literature Review

Enhancing Urban Resilience through Green Infrastructure: Kathmandu Case Study

1. Urban Flooding and Hydrological Changes in Kathmandu Valley

Rapid urbanization in South Asian cities has significantly altered natural hydrological processes, leading to increased surface runoff and heightened flood risk. Kathmandu Valley represents a critical example of these challenges due to unplanned urban growth, high impervious surface coverage, and monsoon-dominated rainfall patterns.

Several hydrological studies focusing on Kathmandu Valley indicate that urban expansion has reduced infiltration capacity while increasing peak runoff and flood frequency. Peer-reviewed hydrologic–hydraulic modeling studies conducted in Kathmandu demonstrate that land-use change has led to measurable increases in flood inundation extent and peak discharge during extreme rainfall events. These studies consistently show that conversion of agricultural and open land into built-up areas intensifies surface runoff and shortens time-to-peak during monsoon storms.

Rainfall intensity plays a critical role in flood initiation in the Bagmati River corridor. Observational and modeling-based studies on monsoon hydrology in Kathmandu indicate that short-duration, high-intensity rainfall events are primary flood triggers. Although exact threshold values vary across studies, rainfall events exceeding approximately 40–60 mm within a 24-hour period are commonly associated with localized urban flooding and drainage system exceedance. These findings provide a scientifically reasonable basis for adopting rainfall thresholds around 50 mm/24 hrs in preliminary flood simulations.

Overall, the literature confirms that Kathmandu's flood vulnerability is strongly driven by urbanization-induced hydrological change, making it a suitable case for evaluating green infrastructure-based mitigation strategies.

2. Green Infrastructure and Nature-Based Solutions for Urban Flood Mitigation

Green infrastructure (GI), also referred to as nature-based solutions (NBS), has gained global recognition as an effective approach for mitigating urban flood risk while providing co-benefits such as sediment control, ecosystem enhancement, and climate adaptation. International development organizations and academic literature consistently emphasize GI as a cost-effective and sustainable alternative to conventional grey infrastructure.

Guidelines and toolkits published by the Asian Development Bank (ADB) and similar international institutions highlight the applicability of green infrastructure in tropical and monsoon-affected cities across South Asia. These documents emphasize interventions such as permeable pavements, bio-swales, rain gardens, and urban green corridors. Performance assessments across multiple climatic regions indicate that permeable pavements can reduce surface runoff by approximately 30–60%, depending on soil conditions, pavement design, and maintenance practices.

However, maintenance remains a critical factor influencing long-term performance. Studies and technical guidance reports note that sediment accumulation can significantly reduce infiltration efficiency over time. Without periodic maintenance such as pressure washing or surface cleaning, infiltration capacity may decline by approximately 10–20% annually, particularly in dust- and sediment-rich urban environments like Kathmandu. This finding underscores the importance of integrating maintenance assumptions into performance modeling of green infrastructure.

The reviewed literature strongly supports the integration of green infrastructure into urban drainage systems, especially in cities facing land constraints and climate variability.

3. Sedimentation, Drainage Efficiency, and Bio-Swale Performance

Sedimentation is a major contributor to reduced drainage efficiency in urban flood management systems. South Asian cities experience high sediment loads due to construction

activities, unpaved surfaces, and upstream erosion. As a result, conventional drainage networks often suffer from clogging and reduced conveyance capacity.

Research on nature-based solutions for flood risk management in South Asia indicates that bio-swales and vegetated infiltration systems play a dual role: enhancing infiltration while trapping sediments before they enter primary drainage channels. Empirical and modeling studies conducted in silt-dominated soils show that bio-swales typically exhibit infiltration rates ranging from approximately 2.5 cm/hr to 4.0 cm/hr, depending on soil composition and vegetation density.

Additionally, green infrastructure elements have been reported to trap up to 60–70% of suspended solids, thereby reducing sediment transport into downstream drainage pipes and river systems. This sediment retention capacity is particularly relevant for Kathmandu, where drainage blockages caused by sediment deposition are a recurring issue during monsoon seasons.

These findings provide essential parameter ranges for computational modeling and structural design, supporting the integration of bio-swales and sediment-control measures into climate-smart urban drainage frameworks.

4. Relevance to Urban Resilience and Climate-Smart Design

The reviewed literature collectively highlights the necessity of transitioning from purely grey infrastructure toward integrated green-grey systems to enhance urban resilience. In the context of Kathmandu, green infrastructure offers a scalable and adaptable solution to address flooding, sedimentation, and climate variability without requiring extensive land acquisition.

From a modeling and design perspective, validated parameter ranges derived from literature—such as runoff coefficients, rainfall thresholds, infiltration rates, and sediment trapping efficiencies—enable realistic simulation of pre- and post-intervention scenarios. These parameters support interdisciplinary collaboration between hydrological modeling, structural design, and policy evaluation, aligning closely with the objectives of the Erasmus Mundus MUrCS programme.

5. Summary of Literature Gaps

Despite the growing body of research on green infrastructure and urban flooding, several gaps remain relevant to Kathmandu and similar South Asian cities:

- Limited city-specific calibration of green infrastructure performance parameters.
- Insufficient integration of sediment dynamics into urban flood modeling.
- Lack of interdisciplinary frameworks linking engineering simulations with urban policy implementation.

Addressing these gaps through a Kathmandu-focused case study contributes both scientific and practical value to climate-resilient urban development research.